

‘Two kinds of truths’

Like Descartes, Gottfried Leibniz contributed as much to mathematics as philosophy. His stance was essentially rationalist, but although he believed that knowledge should be accessible by reasoning, he recognized that rational faculties are imperfect and some knowledge must come from the external world. Leibniz identified ‘two kinds of truths: truths of reasoning and truths of fact’. We can establish the truth of a statement such as ‘all bachelors are unmarried’ by analysing its terms without reference to facts; but to explore the truth of ‘bachelors live longer than married men’ we would need to look to the facts of the matter. The difference between these two types of statement, referred to by later philosophers as analytic and synthetic respectively, becomes particularly clear if we deny them: a truth of reasoning (a ‘necessary’ truth), such as $2 + 2 = 4$, is impossible to deny without logical contradiction, whereas denial of a truth of fact (a ‘contingent’ truth), such as ‘water boils at 100°C ’, simply creates another synthetic statement, not a logical impossibility.

NECESSARY TRUTH:



All squares have
four sides.

CONTINGENT TRUTH:



All cows have
four legs.